

METHODS AND RESULTS ADDENDUM

Learning measurement and QFD-linked analysis

Stage 3B Pro • Initial data batch • Prepared for Sidney K. T. Tam

Bottom line: a reproducible analysis is possible now. The records show immediate game-task performance and an auditable adaptation process, but do not establish durable learning. QFD supports evidence-to-design translation; it is not a statistical test of learning.

1. Purpose and scope

This addendum applies the methods named in the research development record—Quality Function Deployment (QFD), SERVQUAL-inspired experience analysis and Python—to the ten uploaded Batch 01 game exports. The user’s term “QDF” is interpreted as QFD because that is the term explicitly recorded in the August 2024 note and the thesis. The original documents and game files have not been amended (Tam, 2024; Tam, 2026, Table 5.3).

Evidence layer	What was computed	Current finding
Immediate task performance	Opening/closing obstacle-clearance differences	First complete visit per code: 97.5% to 95.0%; mean change -2.5 percentage points.
Experience	Original questionnaire distributions	First-visit challenge-fit responses all 4/5; perceived improvement varies from 1/5 to 5/5.
Adaptation fidelity	Linked decisions and applied settings	Ten recorded decisions match subsequent settings; seven local-rule decisions and three player overrides.
QFD design synthesis	Evidence-to-requirement mapping	Qualitative mapping completed; importance weights were not collected, so no numeric priority ranking is claimed.

Dataset boundary

Ten exports contain 81 run appearances but only 42 distinct runs, after matching stored player code and Run_ID. They represent five stored codes and 12 software-recorded visits. The files retain a researcher-self-test classification; five independent people have not been verified. The older PKY3B7 v4.0 test and all original Stage 1, Stage 2 and Stage 3A study records are excluded (sources S01–S10).

All statistical choices in this addendum are retrospective and exploratory. Numeric tests below are conditional calculations, not confirmatory evidence. The analysis does not validate FIAS, clinical balance, eye-leg coordination, an AI model or the effectiveness of the complete intervention.

2. What the existing tools can measure

The earlier January 2025 review used the heading “QFD Statistical Modeling”. The present analysis preserves QFD’s place in the research, but separates its engineering role from hypothesis testing. The V38 methodological note already describes Table 5.3 as a qualitative, unweighted requirements map, not an auditable weighted QFD deployment.

Tool / framework	Appropriate role here	Boundary
QFD / House of Quality	Translate interpreted player needs into technical requirements and verification measures.	Designed relationships are analyst judgements, not measured effect sizes or proof that learning occurred.
SERVQUAL-inspired items	Describe the player’s reported experience using the questions actually administered.	The six custom items are not full SERVQUAL. No matched expectation ratings exist, so P–E gaps cannot be calculated.
TAM / Octalysis	Explain why challenge fit, progress, choice and continued use matter to the design.	A design linkage does not create validated TAM or eight-drive scale scores in this batch.
Paired descriptive statistics	Calculate how each code performed on the opening and closing checks.	Nominal conditions can be compared; alternate-course equivalence and true person identity are unresolved.
Paired t / exact tests	Show the uncertainty and direction of the available paired differences.	Conditional illustrations only: small sample, bounded scores, unverified independence and no control group.
Learning assessment	Link a specified skill to observable task evidence and later reassessment.	No adequately controlled retention or transfer measurement is established here.

QFD is used as a requirements-translation method, following ASQ’s description of connecting customer expectations to product characteristics. SERVQUAL concerns service-quality perceptions relative to expectations; an experience rating is not itself a skill assessment (ASQ, n.d.; Parasuraman et al., 1988).

Separate the learning constructs

The candidate target for this game is task-specific steering and route selection, not clinical balance. Immediate performance is observable now. Relatively lasting skill development requires a credible retention or transfer assessment; temporary support or easier conditions can change performance without establishing that development (Soderstrom & Bjork, 2015).

A defensible assessment starts with a claim about the intended competence, specifies the observations that would support it, and designs tasks that can supply those observations. This is the useful connection to embedded game assessment, rather than treating any exported score as a learning measure (Shute & Ventura, 2013).

3. Reconstructed performance comparisons

The candidate main indicator is obstacle-clearance rate. It is recalculated from counts rather than accepted uncritically from a displayed score: $\text{clearance (\%)} = 100 \times (\text{obstacles faced} - \text{obstacles hit}) / \text{obstacles faced}$. Within-visit change is closing clearance minus opening clearance, in percentage points.

Code / visit	Opening	Closing	Change	Course forms
P8P2Z9 / 1	100%	90%	-10 pp	7043 → 7043m
PT8FJC / 1	95%	100%	+5 pp	7043 → 7043m
PEVNZ8 / 1	95%	90%	-5 pp	7043 → 7043m
PZTPCG / 1	100%	100%	+0 pp	7043 → 7043m
P8P2Z9 / 4	80%	90%	+10 pp	7051m → 7051

Each complete pair has 20 obstacles, a 30-second planned duration, $\text{pace} \times 0.85$, no route guide and the same input mode within the pair. The first four pairs are from v4.1; the later P8P2Z9 Visit 4 pair is from v4.1.1. Course forms differ within each pair. Mirroring and matching nominal settings do not independently establish equal difficulty (S03, S08–S10).

Retrospective selection rule

The main summary uses the earliest complete paired visit for each stored code: P8P2Z9, PT8FJC, PEVNZ8 and PZTPCG. This avoids counting P8P2Z9 twice as though it were two people. PU5K58 has no complete opening/closing pair and is not assigned a fabricated post-score. Its other records remain in the logbook.

Ceiling and interpretability

Two of the four selected opening checks are already at 100%, and their mean is 97.5%. The obstacle-clearance task has little headroom in these records. One obstacle represents five percentage points. This motivates checking measurement sensitivity; it is not a reason to discard high baselines or selectively replace the outcome with a more favourable score.

Supplementary indicator, first pair/code	Mean change	Interpretation
Coins collected	-0.25 coins	Not automatically worse: safer coin selection could reduce collection.
Time at the edge	-3.21 percentage points	Strategy context, not a validated skill score.
Inherited game composite	-0.90 points	Supplementary game metric only; explicitly not clinical balance.

No complete pair exceeds the earlier prototype’s 50% edge-time screening threshold. Reaction time cannot be reconstructed as a valid latency measure from these summary fields; elapsed obstacle-event time is not the same as stimulus-to-response latency.

4. Statistics, uncertainty and sensitivity

The following confidence interval and p-values assume independent comparable code-level pairs and the relevant sampling/distributional conditions. Those assumptions are not verified. These calculations must not be presented as a completed population-level learning trial.

Statistic: first complete pair per code	Result
Number of codes with a complete pair	4
Opening mean → closing mean	97.50% → 95.00%
Mean change; median change	-2.50 pp; -2.50 pp
SD of paired differences	6.45 pp
Conditional 95% t confidence interval	-12.77 to +7.77 pp
Paired t test	$t(3) = -0.775$; two-sided $p = .495$
Paired standardised change, dz	-0.387
Exact sign test	1 improvement, 2 declines, 1 tie; $p = 1.000$
Exact Wilcoxon signed-rank calculation	$W = 1.5$; two-sided $p = .750$

For differences d , the paired statistic is $\text{mean}(d) / [\text{SD}(d)/\sqrt{n}]$. The interval is $\text{mean}(d) \pm t(.975, n-1) \times \text{SD}(d)/\sqrt{n}$; $dz = \text{mean}(d)/\text{SD}(d)$. These calculations follow the paired-observation approach (NIST/SEMATECH, n.d.).

The Wilcoxon calculation omits the zero difference, assigns midranks to tied absolute differences and enumerates all eight sign allocations for the three non-zero differences. It is independently checked against SciPy's exhaustive permutation implementation. An automatic normal approximation is not used. The signed-rank interpretation additionally assumes a symmetric difference distribution; the sign test discards magnitudes (SciPy contributors, n.d.).

Repeated-visit sensitivity

Across all five observed visit-pairs, the mean change is 0.0 pp, with two improvements, two declines and one tie. This is descriptive, because one code supplies two pairs. An alternative that averages changes within each code before averaging across codes also yields 0.0 pp: the four code means are 0, +5, -5 and 0 pp. That answers a different question from first-visit change; it is not a reason to select whichever result is preferable.

What these results do not mean

There is no consistent positive performance signal in this batch. Neither $p = .495$ nor $p = .750$ establishes the absence of learning, equivalence, or ineffectiveness. The interval is wide and the measurement has a ceiling. Conversely, fitting a more elaborate model cannot supply missing identity, comparable tasks or a causal contrast. No multivariable regression, mediation, random forest or structural-equation model is fitted.

5. Player experience and adaptation

The original six questionnaire items are analysed separately, using the first recorded visit per code to avoid giving frequent visitors extra weight. Missing responses remain missing. The table preserves ordinal medians and ranges; the workbook also contains complete frequency counts.

Item	Answered / missing	Median	Range
D1 Coordination challenge	5 / 0	4	3-5
D2 Willingness to play regularly	5 / 0	4	3-4
F1 Game matched ability	5 / 0	4	4-4
F2 Progress motivated continuation	5 / 0	3	3-5
C1 Felt getting better	4 / 1	4	1-5
T1 Felt tested or judged	5 / 0	3	2-3

P8P2Z9 and PEVNZ8 both selected C1 = 4 at Visit 1, while their clearance changes were -10 and -5 pp respectively. This is a within-record contrast between perceived improvement and the selected performance measure, not evidence that the players' reports are wrong. F1 is constant at 4 across all five first responses, so its correlation with performance is undefined.

C1 is not a validated self-efficacy score. T1 is also not a positive-outcome item: higher values represent more perceived judgement. The six items span different constructs, so no unvalidated total score, Cronbach-alpha claim or synthetic "learning percentage" is produced. No full SERVQUAL expectation-perception gap is estimable.

Process fidelity, not an effectiveness estimate

All ten archived decision chains match their next-run settings, but they occur within only two stored codes. Seven final decisions are local-rule outputs and three are player overrides. No cloud configuration or live external model request is recorded, and the free-text comments are empty. Therefore this batch does not test Jev language reasoning. The observed 10/10 agreement is retained as a software-process count, not analysed as ten independent participants (S03, S10).

Completeness and participation

Seven of the twelve visits contain practice. Five of those seven include both checks; two do not. Five visits have no practice. Two sequential visits in one code begin only about 9.5 and 3.4 seconds after the preceding visit ended. Software visit counts therefore do not define durable voluntary return. A later-day code record is present, but a planned retention test without preceding re-warmup and with a verified comparison task is not established.

6. QFD: from evidence to requirements

The completed QFD-style matrix is in the workbook. Its rows distinguish player reports from analyst-inferred usability and measurement requirements. Strong or moderate links are proposed design judgements, not empirical correlation coefficients. This follows the qualitative status already stated in V38 rather than inventing customer-importance weights retrospectively.

Requirement	Evidence	Proposed engineering response
Appropriate challenge	First-visit F1 ratings all 4; hard/easy feedback varies during practice.	Separate directional speed feedback; assess fit without equating ease with learning.
Credible personal progress	C1 varies 1–5; fixed-course changes are mixed.	Keep performance, perceived improvement and durable learning visibly distinct.
Understandable support	Ten linked decisions; no free-text interpretation tested.	Log the actual advice, relevant next action and later unassisted performance.
Simple visit navigation	Five visits contain no practice; two practice visits lack closing checks.	Clarify optional steps and finishing routes; do not force continued play.
Separate player records	Five codes, but human identities unverified.	Confirm consistent code use and returning-player history before inference.
Voluntary continuation	D2 median 4; T1 median 3; short return gaps.	Keep choice, pause and finish; distinguish immediate replay from later return.
Interpretable learning assessment	Nominally comparable checks; form equivalence unresolved.	Evaluate course forms and a delayed comparison in a separate later protocol.

Why no numerical QFD ranking is reported

A weighted QFD score could be defined as $P(j) = \sum w(i) \times r(i,j)$, where w is a documented importance weight and r is a justified design relationship. Neither quantity is an observed learning effect. No elicited importance weights were found in these exports. The optional QFD_Template sheet therefore keeps weights and numeric relationships blank and returns PENDING until all required inputs and provenance are supplied.

The template offers conventional 0/1/3/9 relationship coding only as a documented future design exercise; it does not convert the qualitative matrix into an empirical finding. Satisfaction scores, missing expectations and missing importance weights are not interchangeable. A target of 5 on a rating scale has not been substituted for an unmeasured expectation.

The contribution available now is organisational/design learning: evidence exposes where the intended process needs revision. That can be documented without claiming that the player acquired new knowledge or that the system trained a model.

7. Research-review wording and next use

Suggested results paragraph

“A retrospective exploratory analysis of the initial Stage 3B Pro exports identified five complete opening-closing game-performance pairs across four stored player codes. Using the earliest complete pair per code, mean obstacle clearance changed from 97.5% to 95.0% (mean difference -2.5 percentage points). Conditional paired calculations gave a 95% interval of -12.77 to +7.77 points, $t(3) = -0.775$, $p = .495$; exact signed-rank and sign-test calculations were also inconclusive. Because participant identity, independent sampling and course equivalence were not established, these statistics are not treated as confirmatory evidence. QFD-style mapping translated the mixed performance, experience and process findings into candidate engineering requirements. No durable learning, clinical improvement or live Jev effectiveness is claimed.”

What is useful before thesis closure

Retain this as a separate development addendum, not as a replacement for the original thesis's Stage 1 survey or the frozen Stage 3B study. Complete the source-to-person clarification and preserve which records are self-tests. Do not pool these phone-game metrics with Stage-2 clinical FRA indices or recreate missing earlier measures.

For the present review, show the paired table, the uncertainty, the perception-performance distinction and the QFD mapping. Do not report the exploratory p-values without their assumptions. This analysis does not require reopening the thesis scope or starting a new Pro participant trial.

A later learning evaluation

A later, separately approved design could define one target skill, fix or empirically equate the assessment conditions, establish a baseline before practice and examine an unassisted delayed test. Delayed performance should be assessed before a new warmup when the aim is retention, or any re-warmup exposure must be explicitly part of the estimand. Transfer requires an appropriately different task. The interval, task, primary outcome and meaningful-change criterion should be specified before collecting the next batch.

Repeated-measures modelling becomes a candidate only with sufficient verified people and repeated comparable tasks. A model could relate obstacle-level success to practice exposure, challenge and support with within-person dependence represented. It would still estimate an association unless a suitable comparison design identifies the adaptation effect. No such model is fitted to this batch.

Reproducibility and checks

The companion workbook contains 12 sheets. The package retains the ten original exports, canonical tables, independent Python re-extraction and results.json. Verification confirms source checksums, 81-to-42 run deduplication, source-count clearance rates, matching canonical Run_ID sets, two implementations of the paired t calculation, and exhaustive signed-rank enumeration against SciPy. These are internal verification checks, not verification of the individuals who played.

Analysis environment: Python 3.13.5; NumPy 2.3.5; SciPy 1.17.0. No network or model call is made by the analysis script.

References and source records

[R1] Tam, S. K. T. (2024). my research update aug 2024 [research development note].

[R2] Tam, S. K. T. (2026). Full version V38 EngD thesis submission draft, Table 5.3 / Section 5.5A.

[R3] American Society for Quality. (n.d.). What is Quality Function Deployment (QFD)?

[R4] Parasuraman, A., Zeithaml, V. A., & Berry, L. L. (1988). SERVQUAL: A multiple-item scale for measuring consumer perceptions of service quality. *Journal of Retailing*, 64(1), 12-40.

[R5] NIST/SEMATECH. (n.d.). e-Handbook of Statistical Methods, Section 7.3.1.1, Analysis of paired observations.

[R6] SciPy contributors. (n.d.). `scipy.stats.ttest_rel`; `binomtest`; `wilcoxon` [software documentation].

[R7] Soderstrom, N. C., & Bjork, R. A. (2015). Learning versus performance: An integrative review. *Perspectives on Psychological Science*, 10(2), 176-199. DOI: 10.1177/1745691615569000.

[R8] Shute, V. J., & Ventura, M. (2013). *Stealth assessment: Measuring and supporting learning in video games*. MIT Press.

Online methodological sources were checked on 20 September 2026. URLs and analytical roles are supplied in the workbook Sources sheet. R1-R2 are historical research documents, not evidence of current intervention effectiveness.

Batch 01 empirical source files

S01 Alpine3BPRO_PRO-SELF_PU5K58_v2_202609191732.json

S02 Alpine3BPRO_PRO-SELF_PU5K58_v1_202609191608.json

S03 Alpine3BPRO_PRO-SELF_P8P2Z9_v4_202609200321.json

S04 Alpine3BPRO_PRO-SELF_P8P2Z9_v2_202609191523 2.json

S05 Alpine3BPRO_PRO-SELF_P8P2Z9_v3_202609191604.json

S06 Alpine3BPRO_PRO-SELF_PU5K58_v3_202609191735.json

S07 Alpine3BPRO_PRO-SELF_P8P2Z9_v2_202609191523.json

S08 share5214078768679196894.csv

S09 Alpine3BPRO_PRO-SELF_PZTPCG_v1_202609191427.csv

S10 Alpine3BPRO_PRO-SELF_PT8FJC_v1_202609191040.csv

Source IDs follow the preceding Batch 01 logbook. Duplicate/cumulative exports were retained for provenance, not added as independent observations. All ten unchanged raw files and SHA-256 checksums are in the package. The earlier logbook was used as the reconciliation reference; all run counts and paired measures were independently re-extracted here.